QUASI-FINITENESS OVER A CODIMENSION ZERO POINT: THREE EQUIVALENT FORMULATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5098-5108, immediately following the statement at 5084-5096.

Editorial note. Two cross-chapter pointers (Properties of Spaces, Morphisms) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Etale-local detection of quasi-finiteness. standard etale-locality of quasi-finiteness invoked at proof lines 5102-5104 (f quasi-finite at x iff U -> V quasi-finite at u for the etale charts). Lets the spaces-side statement be moved to the scheme charts U -> V.
- Codimension zero points under etale maps. Properties of Spaces, Lemma spaces-properties-lemma-codimension-0-points, cited at proof lines 5104-5106. x in X^0 iff u is the generic point of an irreducible component of U – aligns the point sets across the charts.
- Scheme-side equivalence over generic points. Morphisms, Lemma morphisms-lemma-quasi-finiteness-over-generic-point, cited at proof lines 5107-5108. The imported scheme statement that the lemma reduces to.

Source reference. Original reference 0BAH: focused-seeds/geometry/S9-decent-quasi-finiteness-over-generic-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAQ: focused-seeds/geometry/S9-decent-quasi-finiteness-over-generic-point/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type. Let $X^0 \subset |X|$, resp. $Y^0 \subset |Y|$ denote the set of codimension 0 points of X, resp. Y. Let $y \in Y^0$. The following are equivalent*

(1) *$f^{-1}(\{y\}) \subset X^0$,*
(2) *f is quasi-finite at all points lying over y,*
(3) *f is quasi-finite at all $x \in X^0$ lying over y.*

Proof. Let V be a scheme and let $V \to Y$ be a surjective étale morphism. Let U be a scheme and let $U \to V \times_Y X$ be a surjective étale morphism. Then f is

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quasi-finite at the image x of a point $u \in U$ if and only if $U \to V$ is quasi-finite at u. Moreover, $x \in X^0$ if and only if u is the generic point of an irreducible component of U (Properties of Spaces, Lemma 0BAQ). Thus the lemma reduces to the case of the morphism $U \to V$, i.e., to Morphisms, Lemma 0BAH. $\square$